Empirical Results Report

Miratrix

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Introduction

This report generates the results reported in our paper (but not the figures—see the script `final_paper_figures.R` for those).

Some statistics on the studies

Our count of studies is the following:

	Blocked	Unblocked
Curated	11	0
Public	15	23

Our count of outcomes is the following:

	Blocked	Unblocked
Curated	28	0
Public	22	38

Some statistics about the studies across the 50 outcomes:

statistic	mean	median	sd	min	Q10	Q90	max
J	483.74	56.50	1637.23	16.00	30.00	495.00	8375.00
Jcv	0.40	0.33	0.30	0.00	0.09	0.85	1.00
K	35.16	16.00	65.39	5.00	5.00	85.30	389.00
R2.1	0.21	0.17	0.17	0.00	0.02	0.44	0.58
R2.2	0.43	0.37	0.32	0.00	0.06	0.83	0.95
R_ate	0.06	0.04	0.06	0.00	0.01	0.14	0.30
R_se	1.33	1.25	0.38	1.04	1.06	1.63	3.46
block_ICC	0.09	0.07	0.10	0.00	0.00	0.21	0.39
cluster_ICC	0.09	0.09	0.06	0.00	0.01	0.15	0.28
n	7761.36	2148.50	15702.56	409.00	854.40	17499.30	102869.00
nbar	60.41	25.37	110.50	1.74	10.19	86.35	588.04
ncv	0.41	0.34	0.21	0.01	0.21	0.60	1.35
p.tx	0.55	0.50	0.10	0.40	0.49	0.74	0.80

The 10th and 90th percentile number of individuals, clusters and blocks are around: 850 to 17,500 (individuals), 30 to 495 (clusters), and 5 to 85 (blocks).

unit	10%	90%
n	854.4	17499.3
J	30.0	495.0
K	5.0	85.3

Cut-off for "low" degrees of freedom

Low is strictly less than 10

Which studies have singletons and doubletons?

This table lists the number of singletons and doubletons for the Blocked experiments

UniqueID	study_id	num_singletons	num_doubletons
Beacon_S8TGPA	NA	27	13
Beacon_S8TOTCR	NA	27	13
Beacon_SRPRST2	NA	27	13
EMERGE_team2__BB	NA	2	0
EMERGE_eclsb2_BB	NA	2	0
EMERGE_team2__BBSSR	NA	2	0
EMERGE_eclsb2_BBSSR	NA	2	0
CARES_DC_PT_Incred	NA	36	8
CARES_TR_BPI_Incred	NA	36	8
CARES_TR_WRS_Incred	NA	36	8
CARES_DC_EMIT_PTH	NA	36	8
CARES_DC_CST_COMPTOT_PTH	NA	36	8
CARES_TR_SSRS_PTH	NA	36	8
CARES_DC_PT_Tools_Mind	NA	36	8
CARES_TR_WRS_Tools_Mind	NA	36	8
CLC_GRCCSS_POST	NA	14	1
CLC_PCCRERCORE	NA	14	1
MathPD_MTSS	NA	0	1
MathPD_MTSS_FRACTDEC	NA	0	1
MathPD_MTSS_PCTRATIO	NA	0	1
MPC_eclsb2_scale	NA	1	25
MPC_wj_ap2	NA	1	25
PowerTeaching_TSTSSTAND	NA	0	2
Success_S14WJLWI_SCORE	NA	0	4
TennStar_gktmathss	NA	37	87
TennStar_gktreadss	NA	37	87
TRIAD_rema2	NA	2	3
TRIADScale-up_rema2	NA	16	0
jg_crt_2	2	0	0
jg_crt_6	6	769	7
jg_crt_7	7	0	12
jg_crt_8	7	0	12
jg_crt_10	9	0	0
jg_crt_11	9	0	12
jg_crt_12	9	0	12
jg_crt_16	11	0	0
jg_crt_17	12	12	9
jg_crt_18	12	12	9

UniqueID	study_id	num_singletons	num_doubletons
jg_crt_19	12	12	9
jg_crt_23	16	25	5
jg_crt_29	19	0	0
jg_crt_30	19	0	0
jg_crt_31	20	79	13
jg_crt_34	23	0	0
jg_crt_41	27	0	0
jg_crt_52	35	508	0
jg_crt_56	38	5	5
jg_crt_63	42	161	9
jg_crt_68	45	79	13
jg_crt_69	45	79	13

So the number of outcomes that had no singletons is

blocked	Collection	n_outcome	n_no_singletons	n_no_doubletons
Blocked	Curated	28	5	5
Blocked	Public	22	11	8
Unblocked	Public	38	38	38

Studies with few degrees of freedom

UniqueID	cons_df	n	J	K	Jbar	Jcv	ncov.1	ncov.2
TRIADScale-up_rema2	0	409	16	8	2.00	0.00	1	1
jg_crt_52	0	102869	508	254	2.00	0.00	2	2
CARES_DC_PT_Incred	8	1298	52	22	2.36	0.33	1	1
CARES_TR_BPI_Incred	8	1318	52	22	2.36	0.33	1	1
CARES_TR_WRS_Incred	8	1320	52	22	2.36	0.33	1	1
CARES_DC_EMIT_PTH	8	1229	52	22	2.36	0.33	1	1
CARES_DC_CST_COMPTOT_PTH	8	1206	52	22	2.36	0.33	1	1
CARES_TR_SSRS_PTH	8	1276	52	22	2.36	0.33	1	1
CARES_DC_PT_Tools_Mind	8	1268	52	22	2.36	0.33	1	1
CARES_TR_WRS_Tools_Mind	8	1293	52	22	2.36	0.33	1	1
jg_crt_23	9	926	41	16	2.56	0.43	2	2
jg_crt_63	9	8987	179	85	2.11	0.15	4	4
CLC_GRCCSS_POST	10	9557	28	9	3.11	0.78	2	2
CLC_PCCRERCORE	10	16465	28	9	3.11	0.78	2	2
jg_crt_7	16	1386	30	7	4.29	0.18	11	11
jg_crt_8	16	1335	30	7	4.29	0.18	11	11
jg_crt_11	16	2588	30	7	4.29	0.18	12	12
jg_crt_12	16	2612	30	7	4.29	0.18	12	12
TRIAD_rema2	24	1208	38	7	5.43	0.10	1	1
jg_crt_6	24	15993	802	389	2.06	0.42	15	15
MathPD_MTSS	27	2132	39	6	6.50	0.19	1	1
MathPD_MTSS_FRACTDEC	27	2132	39	6	6.50	0.19	1	1
MathPD_MTSS_PCTRATIO	27	2132	39	6	6.50	0.19	1	1
Success_S14WJLWI_SCORE	27	1631	37	5	7.40	0.74	1	1

Some statistics on our estimators

The number of large ranges of SE (2x or more) and ATE (0.10 or more) is the following (*** means yes, blank means no):

```
##              big SE ratio
## big ATE range    ***
##                40   1
##           ***  8   1
```

The number of ATE ranges greater than 0.10 and 0.05

group	measure	n	n_10	big_10	n_05	big_5
all	ate	50	9	0.18	19	0.38
estimand	ate_Cluster	50	3	0.06	11	0.22
estimand	ate_Person	50	3	0.06	8	0.16
noFI	ate_noFI	50	8	0.16	16	0.32
noFI_estimand	ate_Cluster_noFI	50	3	0.06	9	0.18
noFI_estimand	ate_Person_noFI	50	3	0.06	3	0.06

How often Standard errors are large

group	measure	n	n_1.2	big_1.2	n_1.5	big_1.5	n_2	big_2	max
all	se	50	29	0.58	8	0.16	2	0.04	3.46
estimand	se_Cluster	50	20	0.40	3	0.06	1	0.02	3.45
estimand	se_Person	50	17	0.34	5	0.10	0	0.00	1.75
noFI	se_noFI	50	20	0.40	5	0.10	0	0.00	1.71
noFI_estimand	se_Cluster_noFI	50	12	0.24	2	0.04	0	0.00	1.62
noFI_estimand	se_Person_noFI	50	12	0.24	3	0.06	0	0.00	1.56

Stability connected to df

When DF are low, the ratio of the largest to smallest standard error is always greater than 1.5x. When DF are high, only 6 out of 34 outcomes have a ratio above 1.5.

Note: The above claim is no longer true, now that we have fixed the Fully Interacted issue.

low_df	group	measure	n	n_1.2	big_1.2	n_1.5	big_1.5	n_2	big_2
high	all	se	34	13	0.38	2	0.06	1	0.03
high	estimand	se_Cluster	34	7	0.21	1	0.03	0	0.00
high	estimand	se_Person	34	11	0.32	1	0.03	0	0.00
high	noFI	se_noFI	34	7	0.21	2	0.06	0	0.00
high	noFI_estimand	se_Cluster_noFI	34	5	0.15	1	0.03	0	0.00
high	noFI_estimand	se_Person_noFI	34	6	0.18	0	0.00	0	0.00
low	all	se	16	16	1.00	6	0.38	1	0.06
low	estimand	se_Cluster	16	13	0.81	2	0.12	1	0.06
low	estimand	se_Person	16	6	0.38	4	0.25	0	0.00
low	noFI	se_noFI	16	13	0.81	3	0.19	0	0.00
low	noFI_estimand	se_Cluster_noFI	16	7	0.44	1	0.06	0	0.00
low	noFI_estimand	se_Person_noFI	16	6	0.38	3	0.19	0	0.00

Fully interacted estimator notes (now defunct)

To consider how much trouble the fully interacted models are causing, we exclude them and recalculate the ratios (see Figure 5). Doing so dramatically changes the story. For example, the largest estimated standard error is 1.5x the smallest estimated standard error in only 10 percent of cases without the fully-interacted models, down from 48 percent when considering all estimators. The ratio never exceeds 1.7 once the fully-interacted models are excluded.

See above tables.

Even in Figure 3 we see estimated standard errors that are 20 percent different across estimators in over 20 percent of the cases we examined.

See above tables

When are SE estimators defined?

While this model-free approach is appealing in theory, in practice this means design-based standard errors can only be estimated if every block X experimental arm has at least two or more clusters. Only three studies in our sample meet this condition; consequently, the design-based standard errors do not exist for XX out of 50 outcomes when the fully-interacted model (for individual or aggregate data) is implemented. The same limitation applies to heteroskedastic robust standard errors. For most outcomes, there are only 13 estimated standard errors, rather than 19 (See Supplemental Excel Spreadsheet 1).

This table counts the number of undefined estimators within each group given the set of covariates included.

Estimator	n	n_ATE_all	n_ATE_best	n_ATE_none	n_SE_all	n_SE_best	n_SE_none
LRa-FE-db	50	0	0	0	2	0	0
LRa-FE-het	50	0	0	0	8	0	0
LRa-FIcw-db	50	12	2	0	40	34	34
LRa-FIcw-het	50	12	2	0	41	34	34
LRapw-FE-db	50	0	0	0	2	0	0
LRapw-FE-het	50	0	0	0	4	0	0
LRapw-FIpw-db	50	12	2	0	40	34	34
LRapw-FIpw-het	50	12	2	0	40	34	34
LRi-FE-crve	50	0	0	0	0	0	0
LRi-FE-db	50	0	0	0	2	0	0
LRi-FIpw-crve	50	10	0	0	40	34	34
LRi-FIpw-db	50	12	2	0	40	34	34
LRicw-FE-crve	50	0	0	0	0	0	0
LRicw-FE-db	50	0	0	0	2	0	0
LRicw-FIcw-crve	50	10	0	0	40	34	34
LRicw-FIcw-db	50	12	2	0	40	34	34
MLM-FE	50	0	0	0	0	0	0
MLM-FIcw	50	0	0	0	0	0	0
MLM-RE	50	0	0	0	0	0	0

This table counts the number of estimates for each outcome that are undefined (the different columns correspond to adjusting for all covariates, a single best covariate, or no covariates):

UniqueID	n	n_ATE_all	n_ATE_best	n_ATE_none	n_SE_all	n_SE_best	n_SE_none
Beacon_S8TGPA	19	0	0	0	8	8	8
Beacon_S8TOTCR	19	0	0	0	8	8	8

UniqueID	n	n_ATE_all	n_ATE_best	n_ATE_none	n_SE_all	n_SE_best	n_SE_none
Beacon_SRPRST2	19	0	0	0	8	8	8
CARES_DC_CST_COMPUT_PTH	19	0	0	0	8	8	8
CARES_DC_EMIT_PTH	19	0	0	0	8	8	8
CARES_DC_PT_Incred	19	0	0	0	8	8	8
CARES_DC_PT_Tools_Mind	19	0	0	0	8	8	8
CARES_TR_BPI_Incred	19	0	0	0	8	8	8
CARES_TR_SSRS_PTH	19	0	0	0	8	8	8
CARES_TR_WRS_Incred	19	0	0	0	8	8	8
CARES_TR_WRS_Tools_Mind	19	0	0	0	8	8	8
CLC_GRCCSS_POST	19	8	0	0	13	8	8
CLC_PCCRERCORE	19	8	0	0	12	8	8
EMERGE_eclsb2_BB	19	0	0	0	8	8	8
EMERGE_eclsb2_BBSSR	19	0	0	0	8	8	8
EMERGE_team2___BB	19	0	0	0	8	8	8
EMERGE_team2___BBSSR	19	0	0	0	8	8	8
MPC_eclsb2_scale	19	0	0	0	8	8	8
MPC_wj_ap2	19	0	0	0	8	8	8
MathPD_MTSS	19	8	0	0	10	0	0
MathPD_MTSS_FRACTDEC	19	8	0	0	10	0	0
MathPD_MTSS_PCTRATIO	19	8	0	0	9	0	0
PowerTeaching_TSTSSTAND	19	8	0	0	8	0	0
Success_S14WJLWI_SCORE	19	0	0	0	0	0	0
TRIADScale-up_rema2	19	6	6	0	8	8	8
TRIAD_rema2	19	0	0	0	8	8	8
TennStar_gktmathss	19	8	0	0	9	8	8
TennStar_gktreadss	19	8	0	0	9	8	8
jg_crt_10	19	0	0	0	0	0	0
jg_crt_11	19	0	0	0	0	0	0
jg_crt_12	19	0	0	0	0	0	0
jg_crt_16	19	0	0	0	0	0	0
jg_crt_17	19	0	0	0	10	8	8
jg_crt_18	19	0	0	0	9	8	8
jg_crt_19	19	0	0	0	8	8	8
jg_crt_2	19	0	0	0	0	0	0
jg_crt_23	19	0	0	0	8	8	8
jg_crt_29	19	0	0	0	2	0	0
jg_crt_30	19	0	0	0	0	0	0
jg_crt_31	19	0	0	0	8	8	8
jg_crt_34	19	0	0	0	0	0	0
jg_crt_41	19	0	0	0	0	0	0
jg_crt_52	19	6	6	0	8	8	8
jg_crt_56	19	0	0	0	8	8	8
jg_crt_6	19	0	0	0	8	8	8
jg_crt_63	19	0	0	0	8	8	8
jg_crt_68	19	0	0	0	8	8	8
jg_crt_69	19	0	0	0	8	8	8
jg_crt_7	19	8	0	0	8	0	0
jg_crt_8	19	8	0	0	8	0	0

ATE Ranges

Studies with a high range (more than 0.15) of ATE estimates

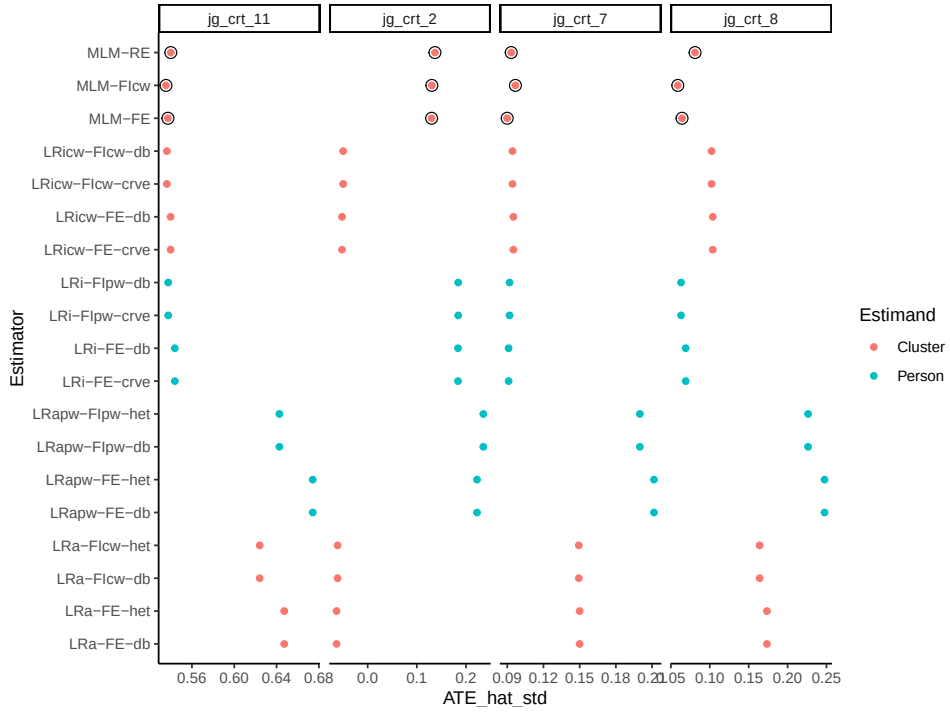
Project	UniqueID	K	J	n	cons_df	R_ate	R_ate_Clu	R_ate_Per
Kim/set_2	jg_crt_2	7	83	2174	69	0.30	0.20	0.05
Kim/set_7	jg_crt_8	7	30	1335	16	0.19	0.11	0.18

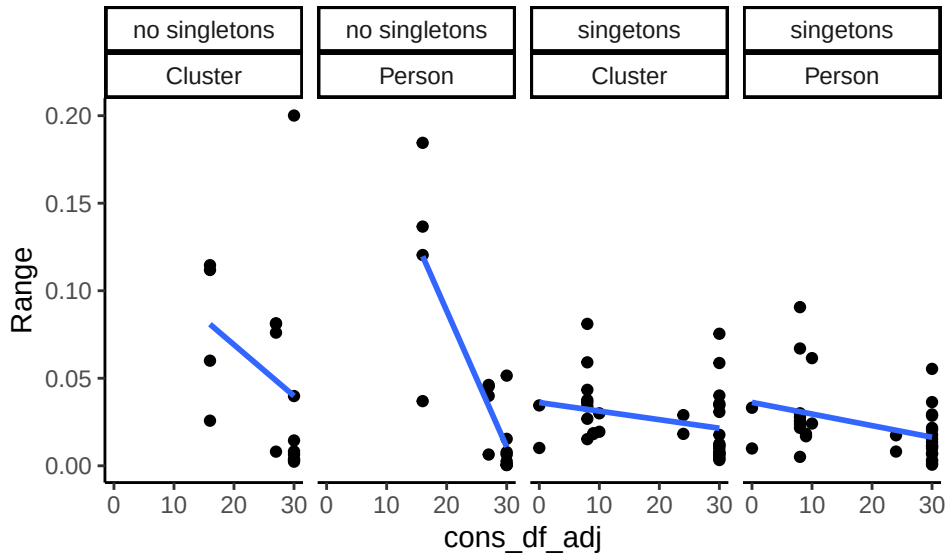
High range within Estimand

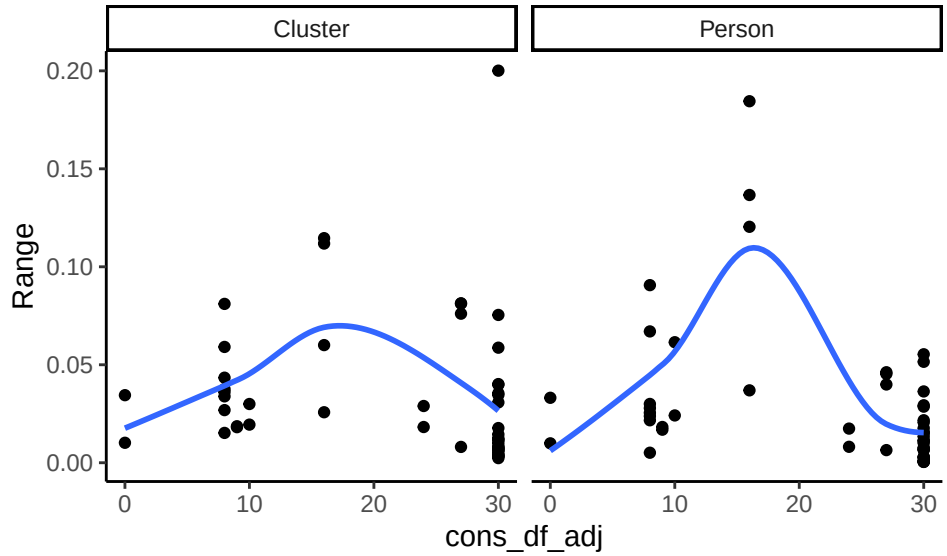
Still a threshold of 0.15.

Project	UniqueID	K	J	n	cons_df	R_ate	R_ate_Clu	R_ate_Per
Kim/set_2	jg_crt_2	7	83	2174	69	0.30	0.20	0.05
Kim/set_7	jg_crt_7	7	30	1386	16	0.12	0.06	0.12
Kim/set_7	jg_crt_8	7	30	1335	16	0.19	0.11	0.18
Kim/set_9	jg_crt_11	7	30	2588	16	0.14	0.11	0.14

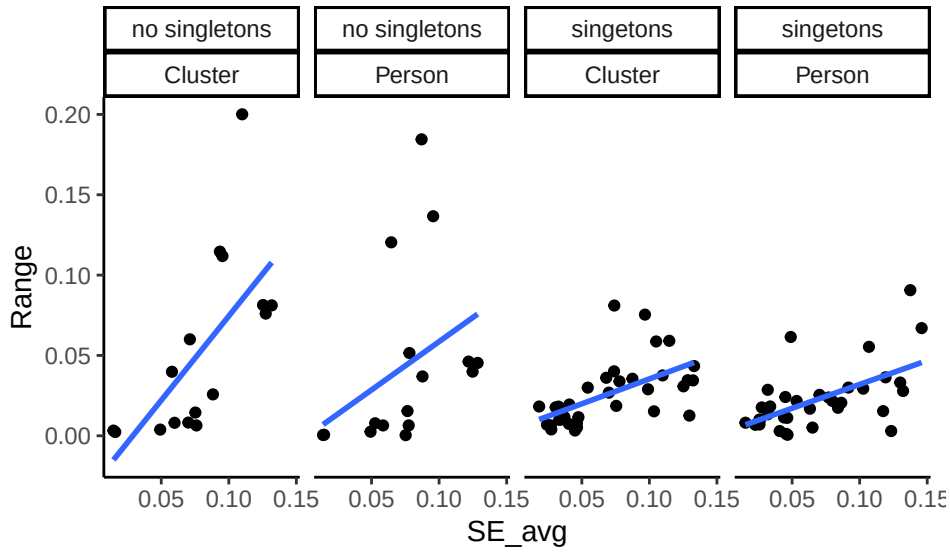
UniqueID	num_singletons	ncv	Jcv	ATE_min	ATE_max	range	SE_avg	n_est	n_def
jg_crt_11	0	0.35	0.18	0.54	0.67	0.14	0.10	19	19
jg_crt_2	0	1.35	0.41	-0.06	0.24	0.30	0.10	19	19
jg_crt_7	0	0.51	0.18	0.09	0.21	0.12	0.07	19	19
jg_crt_8	0	0.52	0.18	0.06	0.25	0.19	0.09	19	19



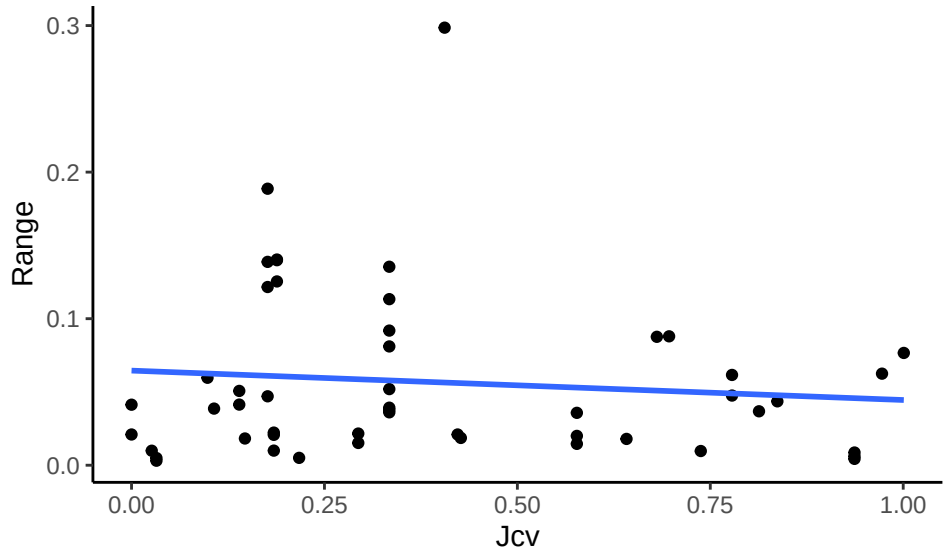


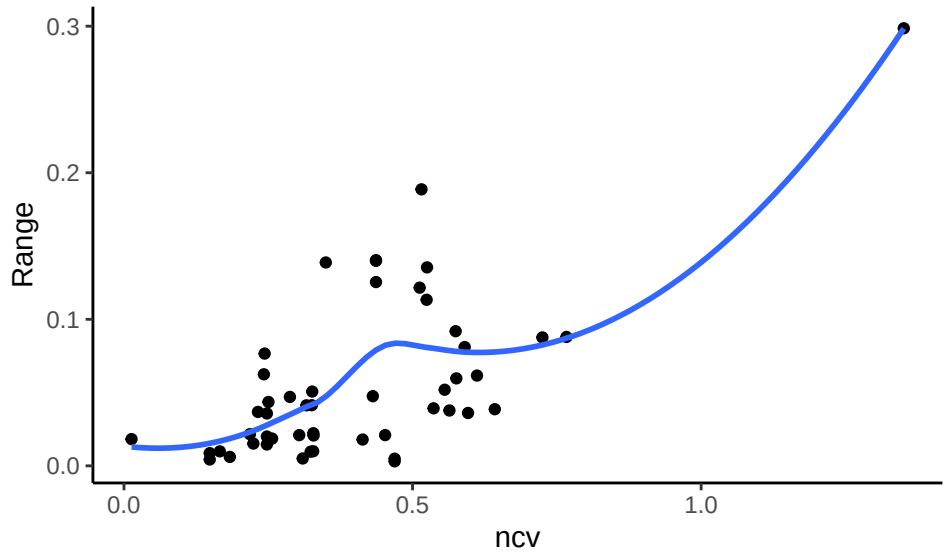


Range does go up as expected with average SE increasing:

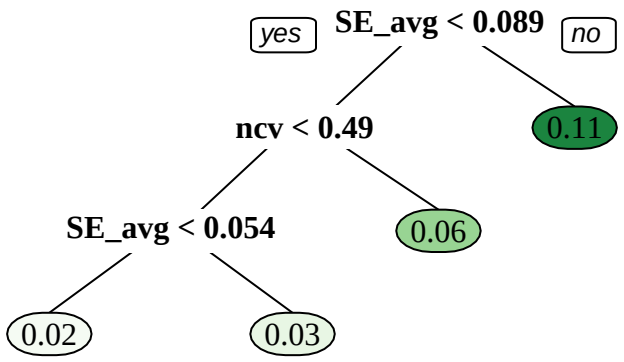


We can see if range depends on variation in block or cluster size:

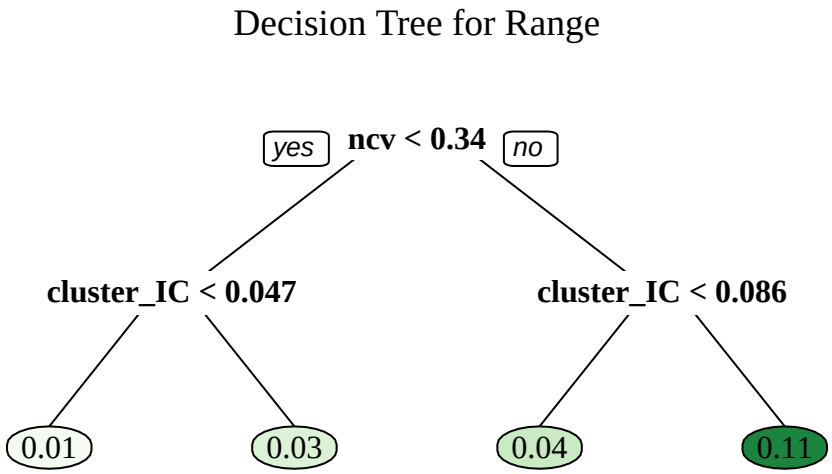




We can use trees to predict what predicts larger range:

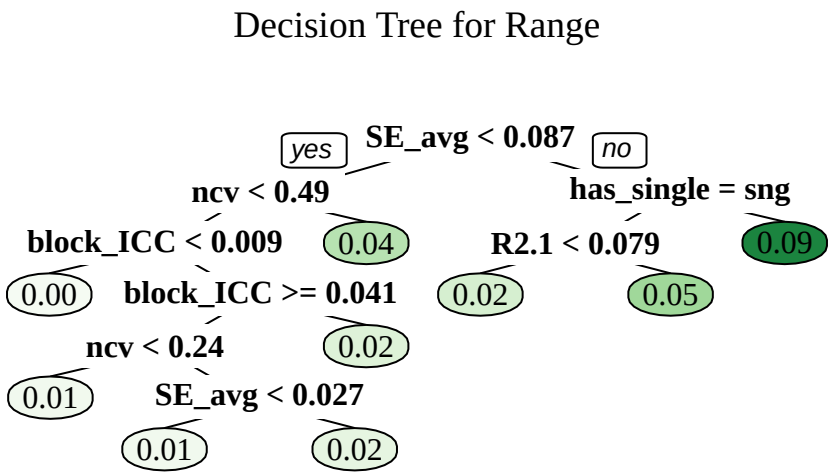


If we drop the average SE we get this:



These trees shows that uncertainty drives larger differences, and also that greater variation in cluster size drives larger differences, especially if there is variation in the cluster means.

If we look within estimand, we see that our datasets with larger R2.1 and no singletons are larger ranges (assuming the SEs are generally large):



Comparing low DF to high DF studies

low_df	avg_rng	sd_rng	Q05	Q95	n	ratio
high	0.05	0.06	0.00	0.14	34	1.00
low	0.07	0.05	0.02	0.15	16	1.36

Standard Errors

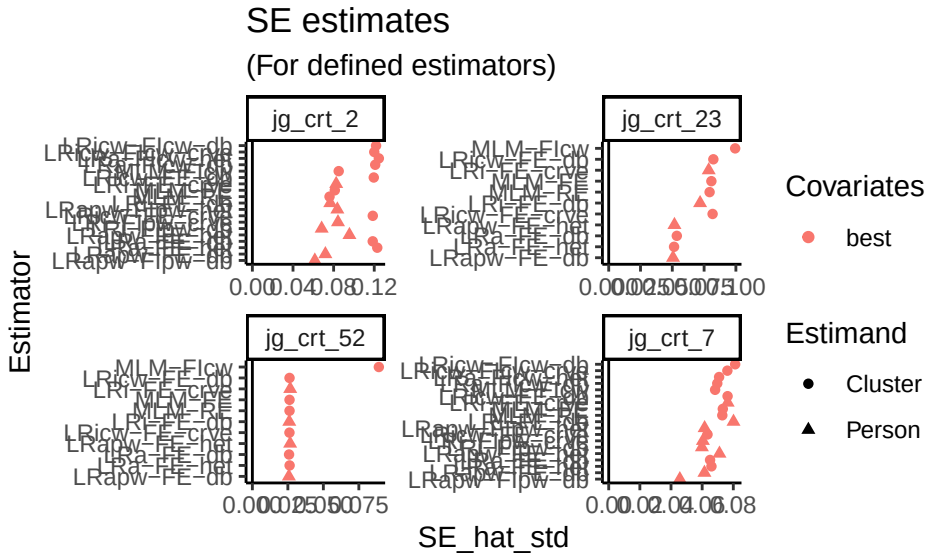
Studies with a high ratio of SE estimates

Here are all the studies with an SE range above 1.75.

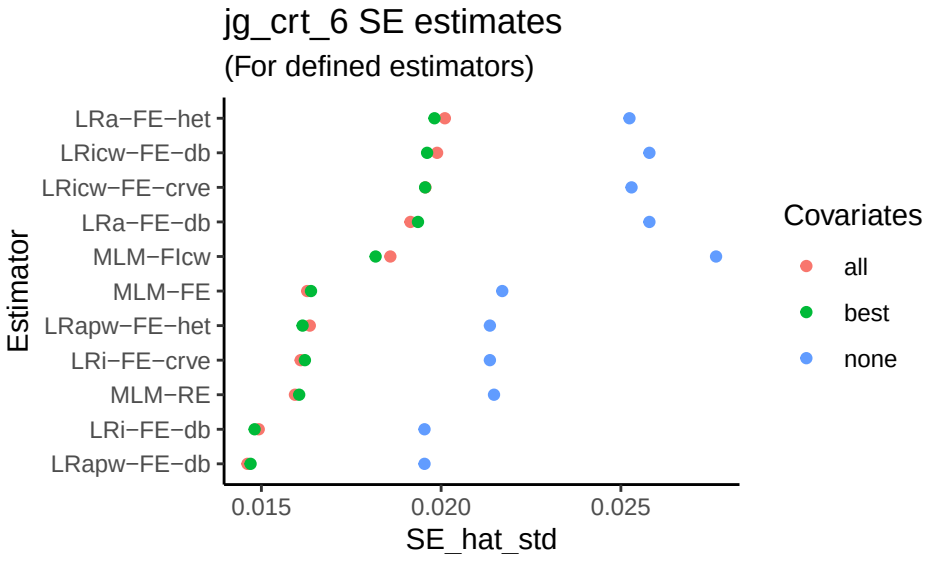
Project	UniqueID	K	J	n	cons_df	R_se	R_se_Clu	R_se_Per
Kim/set_2	jg_crt_2	7	83	2174	69	2.03	1.64	1.55
Bang/set_16	jg_crt_23	16	41	926	9	1.97	1.94	1.56
Abaluck/set_35	jg_crt_52	254	508	102869	0	3.46	3.45	1.04
Kim/set_7	jg_crt_7	7	30	1386	16	1.78	1.28	1.75

UniqueID	cons_df	num_singletons	SE_min	SE_max	n_est	n_def
jg_crt_2	69	0	0.0615	0.1245	19	19
jg_crt_23	9	25	0.0506	0.0994	19	11
jg_crt_52	0	508	0.0257	0.0891	19	11
jg_crt_7	16	0	0.0457	0.0812	19	19

The SEs for the troublesome studies:



Looking at jg_crt_6:



What is the average ratio?

group	Covariates	measure	avg_rng	sd_rng	Q05	Q95	n
all	best	ate	0.06	0.06	0.00	0.14	50
all	best	se	1.33	0.38	1.05	1.88	50
estimand	best	ate_Cluster	0.03	0.04	0.00	0.10	50
estimand	best	ate_Person	0.03	0.04	0.00	0.11	50
estimand	best	se_Cluster	1.25	0.36	1.04	1.56	50
estimand	best	se_Person	1.17	0.18	1.01	1.55	50
noFI	best	ate_noFI	0.05	0.06	0.00	0.14	50
noFI	best	se_noFI	1.22	0.18	1.04	1.60	50
noFI_estimand	best	ate_Cluster_noFI	0.03	0.04	0.00	0.10	50
noFI_estimand	best	ate_Person_noFI	0.02	0.03	0.00	0.09	50
noFI_estimand	best	se_Cluster_noFI	1.15	0.15	1.02	1.45	50
noFI_estimand	best	se_Person_noFI	1.14	0.15	1.00	1.47	50

How often are they defined?

```
##
##        0.578947368421053  1
##   best                34 16
```

How much do SE ratios go down within Estimand?

In the following we look at how much the SE ratios shrink going within estimand.

Are SEs for cluster weight smaller than individual?

```
## lmer(formula = log(SE_hat_std) ~ Estimand + Covariates + has_single +
##     (1 | Estimator) + (1 | UniqueID) + (1 | Estimand:UniqueID),
##     data = combined)
##                     coef.est coef.se
## (Intercept)         -2.66     0.15
## EstimandPerson      -0.03     0.03
```

```
## Covariatesnone       0.17     0.02
## Covariatesall       -0.10     0.02
## has_singlesingetons -0.18     0.18
##
## Error terms:
##  Groups            Name        Std.Dev.
##  Estimand:UniqueID (Intercept) 0.00
##  UniqueID          (Intercept) 0.58
##  Estimator         (Intercept) 0.04
##  Residual                      0.41
## ---
## number of obs: 1965, groups: Estimand:UniqueID, 100; UniqueID, 50; Estimator, 19
## AIC = 2353.2, DIC = 2291.2
## deviance = 2313.2
```

```
##         (Intercept)      EstimandPerson      Covariatesnone       Covariatesall
##           0.06991353          0.97082113          1.18128343          0.90428299
## has_singlesingetons
##           0.83311597
```

Which estimators tend to be most extreme?

Estimator	Estimand	n_def	n_extreme	per
LRa-FIcw-db	Cluster	7	0	0.00
LRi-FIpw-crve	Person	7	0	0.00
MLM-FE	Cluster	25	2	0.08
MLM-RE	Cluster	25	4	0.16
LRa-FE-db	Cluster	25	5	0.20
LRi-FE-crve	Person	25	5	0.20
LRa-FIcw-het	Cluster	7	2	0.29
LRapw-FIpw-het	Person	7	2	0.29
LRi-FIpw-db	Person	7	2	0.29
LRicw-FIcw-crve	Cluster	7	2	0.29
LRicw-FIcw-db	Cluster	7	2	0.29
LRapw-FE-het	Person	25	8	0.32
LRa-FE-het	Cluster	25	9	0.36
LRicw-FE-crve	Cluster	25	9	0.36
LRicw-FE-db	Cluster	25	9	0.36
LRi-FE-db	Person	25	10	0.40
MLM-FIcw	Cluster	25	11	0.44
LRapw-FE-db	Person	25	14	0.56
LRapw-FIpw-db	Person	7	4	0.57

SE ratios vs study characteristics

